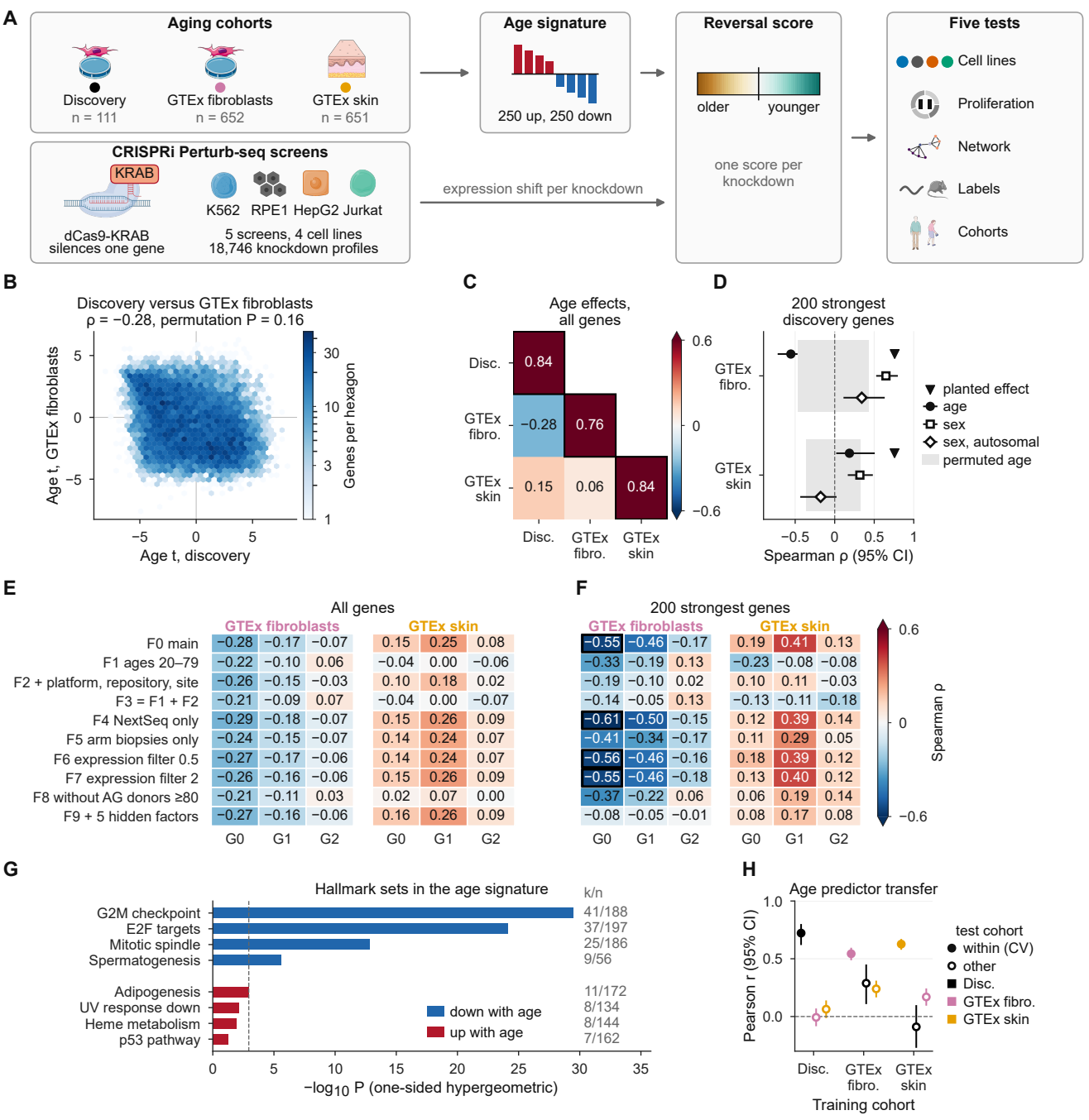

A
Aging cohorts
Discovery
n = 111
GTEx fibroblasts
n = 652
GTEx skin
n = 651
Age signature
250 up, 250 down
Reversal score
older
younger
one score per knockdown
Five tests
Cell lines
Proliferation
Network
Labels
Cohorts
CRISPRi Perturb-seq screens
KRAB
dCas9-KRAB
silences one gene
K562
RPE1
HepG2
Jurkat
5 screens, 4 cell lines
18,746 knockdown profiles
expression shift per knockdown
B
Discovery versus GTEx fibroblasts
ρ = −0.28, permutation P = 0.16
Age t, GTEx fibroblasts
Age t, discovery
Genes per hexagon
C
Age effects, all genes
Disc.
GTEx fibro.
GTEx skin
0.84
−0.28
0.76
0.15
0.06
0.84
D
200 strongest discovery genes
planted effect
age
sex
sex, autosomal
permuted age
Spearman ρ (95% CI)
E
All genes
GTEx fibroblasts
GTEx skin
F0 main
F1 ages 20–79
F2 + platform, repository, site
F3 = F1 + F2
F4 NextSeq only
F5 arm biopsies only
F6 expression filter 0.5
F7 expression filter 2
F8 without AG donors ≥80
F9 + 5 hidden factors
G0
G1
G2
F
200 strongest genes
Spearman ρ
G
Hallmark sets in the age signature
k/n
G2M checkpoint 41/188
E2F targets 37/197
Mitotic spindle 25/186
Spermatogenesis 9/56
Adipogenesis 11/172
UV response down 8/134
Heme metabolism 8/144
p53 pathway 7/162
down with age
up with age
−log10 P (one-sided hypergeometric)
H
Age predictor transfer
Pearson r (95% CI)
Training cohort
test cohort
within (CV)
other
Disc.
GTEx fibro.
GTEx skin